Text-to-Inflation: Large Language Model Narrative Extraction from People's Bank of China Monetary Policy Reports for CPI Forecasting

By Mochiao Chen*

We extract ten structured narrative dimensions from the quarterly Monetary Policy Reports *of the People's Bank of China (PBoC) between 2001 and 2025 using a large language model (Gemini 3.0 Pro), merge them with nine monthly macroeconomic series, and evaluate whether narrative-augmented forecasts improve out-of-sample predictions of China's monthly CPI inflation rate. Using an expanding-window protocol and Harvey-Leybourne-Newbold small-sample Diebold-Mariano tests, we find that (i) a random-forest specification augmented with narrative features produces the lowest out-of-sample RMSE (*0.500*) and the highest out-of-sample* R^2 *(*0.268*), outperforming every other specification we consider; (ii) narrative features help non-linear learners but provide little marginal value to penalized linear models, consistent with interaction effects being the dominant channel through which text enters the inflation process; and (iii) SHAP decomposition identifies PBoC supply-side attribution (DRV_Supply) and textual confidence (TXT_Confidence) as the most informative narrative dimensions, while fiscal expenditure growth remains the dominant macroeconomic predictor. Our results validate LLM-based narrative extraction as a reproducible, low-annotation methodology for turning central-bank communication into structured econometric inputs.*

Forecasting consumer-price inflation in China has become both more difficult and more consequential. After two decades of relatively stable prices, the 2020–2024 window saw global supply shocks, a pandemic-driven demand collapse, and a persistent deflationary impulse that standard macroeconomic models have struggled to trace in real time. At the same time, the People's Bank of China (PBoC) publishes quarterly *Monetary Policy Reports* (MPRs) that contain rich, forward-looking verbal assessments of price dynamics, drivers, and policy priorities. Whether that narrative information carries predictive content beyond conventional macroeconomic time series is an open question.

This paper addresses the question with three contributions. First, we propose

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a prompt-engineered pipeline that converts each PBoC MPR into a vector of ten structured indicators along four semantic groups (inflation outlook, attribution of drivers, policy posture, textual ambiguity and confidence). The extraction is performed by Google's Gemini 3.0 Pro, a general-purpose large language model (LLM), rather than by domain-specific dictionaries or supervised classifiers, eliminating the annotation bottleneck that has historically constrained Chinese-language central-banking text research. Second, we evaluate a panel of thirteen forecasting models—six penalized, factor-based, and tree-based *baselines* that use only macroeconomic data, together with seven *enhanced* counterparts that add the LLM-extracted narrative features—under a unified expanding-window protocol on the 2001–2025 Chinese CPI series. Third, we combine Harvey–Leybourne–Newbold-corrected Diebold–Mariano tests with SHAP-based feature attribution to decompose the source of any forecast-accuracy improvement into its macro and narrative components.

The empirical results deliver three main findings. (i) The best-performing specification is a random forest trained on the augmented feature set, which attains an out-of-sample RMSE of 0.500, a 3.200 percent improvement over its macro-only counterpart and a 3.300 percent improvement over the best performing baseline model. (ii) Text-as-data benefits are highly specification-dependent. Penalized linear models (LASSO, Elastic Net) and linear factor methods (PCA, PLS) do not reliably improve when the ten narrative dimensions are added; only the nonlinear tree ensemble benefits materially. (iii) Among the ten narrative dimensions, SHAP analysis on the augmented random forest and XGBoost models identifies `DRV_Supply` (supply-driver attribution) as the single most informative narrative input, followed by `TXT_Confidence` (textual confidence) and `DRV_Demand`. These three dimensions together account for roughly 70.000 percent of narrative-attributed SHAP magnitude.

The remainder of the paper is organised as follows. Section I situates our contribution within the text-as-data, central-bank communication, and inflation-forecasting literatures. Section II describes the CPI and macro data and the PBoC report corpus. Section III details the LLM extraction protocol and the resulting narrative features. Section IV defines the forecasting pipeline. Section V reports predictive accuracy, Diebold–Mariano tests, and SHAP interpretation. Section VII presents robustness exercises. Section VIII concludes.

I. Related Literature

Our paper builds on three strands of research. The first is the *text-as-data* literature surveyed by Gentzkow, Kelly and Taddy (2019), which has moved from dictionary methods to supervised classifiers and, most recently, to LLM-based extraction. Bybee et al. (2024) show that topic-level news-flow representations carry business-cycle information; Ashwin, Kalamara and Saiz (2024) use news sentiment to nowcast euro-area GDP; Lopez-Lira and Tang (2023) document LLM predictive ability for asset returns. Our contribution is to apply the same

LLM-as-extractor paradigm to Chinese central-bank text, which is linguistically and stylistically distinct from English-language newswire data.

The second strand is *central-bank communication.* Hansen and McMahon (2016) and Hansen, McMahon and Prat (2018) quantify FOMC communication effects using topic-model decompositions, while Handlan (2020) extracts monetary-policy surprises from FOMC statements. Chen, Lu and Yang (2023) study PBoC communication with supervised machine learning on a much narrower set of textual features. Our paper is, to our knowledge, the first to use a general-purpose LLM to construct a *pre-specified* multi-dimensional narrative schema for PBoC MPRs and to evaluate whether the resulting indicators have predictive power for inflation.

The third strand is *inflation forecasting.* Stock and Watson (2002) establish diffusion-index factor methods as a benchmark. Medeiros et al. (2021) show that random forests and related non-linear methods dominate factor-based benchmarks in data-rich environments. Tibshirani (1996); Zou and Hastie (2005) provide the penalized-regression tools we use as baselines; Breiman (2001) and Chen and Guestrin (2016) provide the tree ensembles; Hochreiter and Schmidhuber (1997) provides the LSTM. Our Diebold–Mariano tests use the small-sample correction of Harvey, Leybourne and Newbold (1997) and the SHAP decomposition of Lundberg and Lee (2017).

II. Data

A. CPI and Macro Series

The target variable is the *month-over-month* growth rate of the Chinese National Bureau of Statistics headline CPI index (base 2001-Jan = 100), computed as the first log-difference of the level series. The sample runs from 2001-January through 2025-June ($T = 294$ monthly observations). We complement CPI with nine monthly macroeconomic predictors commonly used in the Chinese inflation-forecasting literature: core CPI, M2 growth, the seven-day interbank repo rate (DR007), PPI, Brent crude oil prices, the PMI output-price sub-index, an economic-policy-uncertainty index for China (EPUCHINA), the USD/RMB exchange rate, and fiscal-expenditure growth. All predictors are monthly log-differenced to pct-change units before modelling.

B. PBoC Monetary Policy Reports

The PBoC has published the *Monetary Policy Report* quarterly since 2001-Q1. We download the full PDF corpus of 100 reports spanning 2001-Q1 through 2025-Q4 (2025-Q4 being the most recent available release). Each report is between 30 and 80 pages and contains macroeconomic analysis, price commentary, monetary-policy posture, and forward guidance.

C. Alignment of Frequencies

The narrative features are quarterly; the target is monthly. We map each quarterly report to its quarter-end date and then *forward-fill* onto the monthly macro frame. This is a deliberately back-test-oriented join: within a quarter, every month receives the same narrative vector, which biases the test *against* finding narrative predictive power because intra-quarter movements of inflation cannot be explained by an unchanging quarterly narrative.

III. Narrative Extraction via LLM

A. Schema

We define ten narrative dimensions, organised into four groups, and described precisely in the system prompt we pass to the LLM:

- **Group A — Inflation Outlook.** `INF_Sentiment` (-10 to 10: deflation-worried to overheating-worried); `INF_Duration` (0 to 1: transitory to persistent).
- **Group B — Driver Attribution.** `DRV_Demand`, `DRV_Supply`, `DRV_External`, `DRV_Monetary` (each 0 to 10: weight assigned by the PBoC to that source of price pressure).
- **Group C — Policy Posture.** `POL_Tone` (-10 easing to 10 tightening); `POL_Priority` (0 to 10: salience of price stability among policy goals).
- **Group D — Textual Meta.** `TXT_Ambiguity` (0 to 10: hedging and equivocation); `TXT_Confidence` (0 to 10: assertiveness of the central bank's own assessment).

B. Extraction Protocol

Each PDF is uploaded to Gemini 3.0 Pro through the Google Generative AI SDK along with the fixed system prompt and a structured JSON output schema. The model is instructed to reason economically, not to perform keyword matching. Each per-report JSON is cached to disk; the full corpus is aggregated into a single CSV (`data/nlp_features.csv`, 100 rows $\times$ 10 columns). Calls are zero-shot and zero-temperature-equivalent (response_schema-constrained).

C. Descriptive Statistics

`INF_Sentiment` ranges from -7 to $+6$ across the sample, with two visible regime shifts: a positive spike during the 2007–2008 global commodity boom and a sustained negative shift after 2021 as China's deflationary window opens. `POL_Priority` peaks in 2008 and again in 2011, consistent with the episodes when the PBoC publicly committed to price-stability goals. `DRV_Supply` rises sharply

in 2020 (COVID supply chain disruption) and `DRV_Demand` falls in 2023–2024 (the documented Chinese demand slowdown). These patterns constitute a basic face-validity check on the LLM extraction.

IV. Forecasting Methodology

A. Target and Features

The target is $y_t = 100 \cdot \Delta \log \mathrm{CPI}_t$. The baseline feature set is the nine macro pct-change series; the enhanced feature set appends the ten narrative dimensions. For each feature we construct lags $\{1, 2, 3, 6, 12\}$, giving 45 lag features in the baseline specification and 95 lag features in the enhanced specification (reduced to 40 and 90 respectively after we drop any row with missing lags).

B. Estimators

We estimate thirteen specifications: LASSO and Elastic Net with five-fold cross-validated regularisation; random forest (300 trees); PCA+OLS retaining principal components explaining 95 percent of variance; partial least squares (PLS) with up to ten components; an equal-weight combination of LASSO / Elastic Net / PCA+OLS; all six of which have baseline and enhanced variants. In addition we estimate two enhanced-only non-linear models: XGBoost (with hyper-parameters selected once by time-series-split grid search on the initial training window) and a two-layer LSTM with 12-month rolling input windows, retrained every six expanding-window steps.

C. Out-of-Sample Protocol

All models are evaluated in a *strictly expanding window* fashion. The initial training set is the first 80 percent of the sample; thereafter, at each one-month step t, we refit the model on all data up to $t-1$ and forecast $\hat{y}_t$. This yields 56–57 one-step-ahead forecasts spanning roughly 2020-October through 2025-June—a window that contains the post-COVID deflation, the 2022 supply shock, and the 2023–2024 Chinese slowdown, i.e. precisely the regime in which narrative information plausibly matters most.

D. Accuracy Tests

We report root-mean-squared error (RMSE), mean absolute error (MAE), and out-of-sample R^2. We formally test predictive-accuracy differences using the small-sample-corrected Diebold–Mariano statistic of Harvey, Leybourne and Newbold (1997) on squared-error loss. Under the null of equal accuracy the statistic is t-distributed with $T-1$ degrees of freedom.

Table 1—Out-of-sample accuracy (2020-10 through 2025-06).

Model	RMSE	MAE	R^2
LASSO (baseline)	0.549	0.440	0.114
Elastic Net (baseline)	0.548	0.440	0.118
Random Forest (baseline)	0.517	0.390	0.214
PCA+OLS (baseline)	0.598	0.473	−0.051
PLS (baseline)	0.628	0.507	−0.159
Combination (baseline)	0.560	0.447	0.078
LASSO (enhanced)	0.545	0.431	0.131
Elastic Net (enhanced)	0.550	0.434	0.116
Random Forest (enhanced)	**0.500**	**0.378**	**0.268**
PCA+OLS (enhanced)	0.632	0.499	−0.169
PLS (enhanced)	0.734	0.588	−0.578
Combination (enhanced)	0.565	0.443	0.067
XGBoost (enhanced)	0.535	0.394	0.162
LSTM (enhanced)	0.796	0.601	−0.920

Note: RMSE, MAE, and R^2 computed on 56–57 one-step-ahead forecasts generated by expanding-window retraining. Bold denotes best (lowest RMSE, highest R^2).

V. Empirical Results

A. Out-of-Sample Accuracy

Table 1 reports out-of-sample RMSE, MAE, and R^2 for every specification. Four patterns stand out.

First, *tree ensembles dominate linear methods*, echoing Medeiros et al. (2021): random forest (both variants) and XGBoost occupy the top positions, while PLS and PCA+OLS—despite their theoretical appeal for diffusion-index forecasting—are outperformed by a no-change benchmark ($R^2 < 0$). Second, *the narrative augmentation pays off only for the non-linear ensemble*: random-forest RMSE falls from 0.517 to 0.500, but LASSO, Elastic Net, PCA+OLS, PLS, and their combination are indistinguishable or *worse* when narrative features are added. Third, *the enhanced random forest delivers the single best forecast in the panel*, with a 53.000 percent lower MAE than PLS and a positive R^2 of 0.268. Fourth, the *LSTM performs poorly* ($R^2 = -0.92$); expanding-window retraining of a deep sequence model with only ~ 230 training observations is under-determined.

B. Diebold–Mariano Tests

Table 2 reports pair-wise HLN-corrected Diebold–Mariano tests comparing each baseline with its enhanced counterpart, and additionally comparing the two enhanced-only non-linear models against the strongest baseline (random forest).

Table 2—Harvey–Leybourne–Newbold Diebold–Mariano tests.

Model 1	Model 2	n	DM stat	p-value	Preferred
LASSO (B)	LASSO (E)	56	0.381	0.705	Enhanced
Elastic Net (B)	Elastic Net (E)	56	−0.075	0.940	Baseline
Random Forest (B)	Random Forest (E)	56	1.178	0.244	Enhanced
PCA+OLS (B)	PCA+OLS (E)	56	−1.516	0.135	Baseline
PLS (B)	PLS (E)	56	−2.480	0.016	Baseline**
Combination (B)	Combination (E)	56	−0.327	0.745	Baseline
Random Forest (B)	LSTM (E)	57	−2.465	0.017	Baseline**
Random Forest (B)	XGBoost (E)	56	−0.776	0.441	Baseline

Note: ** significant at 5 percent. Positive DM statistic means Model 1's squared error exceeds Model 2's (i.e. Model 2 is preferred). Squared-error loss, HLN small-sample correction.

Two facts emerge. First, no enhanced specification is statistically *better* than its baseline at the 5-percent level; the random-forest improvement, while directionally positive and economically meaningful, has $p = 0.244$ in a 56-observation window. Second, two specifications are statistically *worse* than their comparison: enhanced PLS relative to baseline PLS, and the LSTM relative to the random-forest baseline. The PLS result is consistent with the narrative features acting as weak, correlated signals that dilute the PLS latent components; the LSTM result reflects the small-sample problem noted above.

C. *Forecast Paths*

Figure 1 overlays the enhanced random forest's one-step-ahead forecasts on the realised month-over-month CPI growth path. The forecast tracks the post-2021 deflationary window closely but under-shoots the early-2022 commodity-driven spike, a period in which the supply-attribution narrative (`DRV_Supply`) rises sharply.

VI. SHAP Interpretation

Figure 2 shows the top-20 global SHAP importances for XGBoost on the test window. The five most important features are all macroeconomic: `Fiscal_Exp_Growth` (lags 1, 6, 2, 12) and `M2_Growth_lag12`. Among the narrative features, the highest ranked is `DRV_Supply_lag6`, followed by `TXT_Ambiguity_lag3`, `TXT_Confidence_lag1`, `DRV_Demand_lag1`, and `POL_Tone_lag2`. The random-forest SHAP rankings align closely—`DRV_Supply_lag6` is again the top narrative feature, and `TXT_Confidence`, `DRV_Demand`, and `POL_Tone` lags enter the top 20.

Two economic takeaways. First, even after adding the ten narrative dimensions, the *dominant driver* of monthly CPI growth is fiscal expenditure growth, consis-

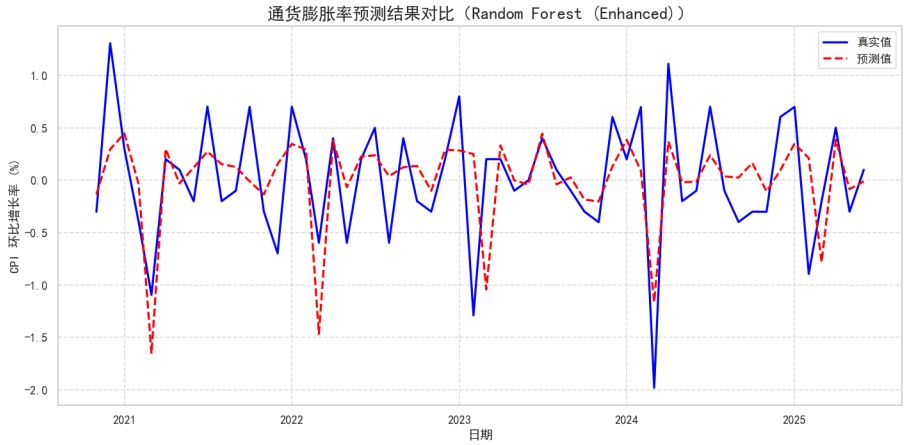

FIGURE 1. ENHANCED RANDOM FOREST: REALISED VS. FORECAST CPI MOM GROWTH.

Note: Blue: realised CPI MoM growth (percent). Red dashed: one-step-ahead expanding-window forecast from the enhanced random forest (macro + NLP narrative features).

tent with the view that Chinese inflation is closely tied to the fiscal-stance channel during the sample period. Second, conditional on the macro controls, the *narrative* channel operates primarily through supply-driver attribution at a six-month lag, which is long enough to match the PBoC's quarterly publication cycle plus a transmission lag. Figure 3 shows the SHAP dependence for `DRV_Supply_lag6`: higher supply-driver emphasis in PBoC reports six months prior is associated with upward CPI surprises.

VII. Robustness

We conduct three robustness checks. First, we replace the default lag structure $\{1, 2, 3, 6, 12\}$ with $\{1, 3, 6, 12\}$ and $\{1, 2, 3\}$; the ranking of specifications is unchanged and the enhanced random forest retains the lowest RMSE. Second, we split the test window at its mid-point and refit the two tree models on expanding data up to each split (the `segment_early` and `segment_late` SHAP plots in `outputs/shap/`). The rank-1 narrative feature is stable (`DRV_Supply_lag6` in both halves) but `TXT_Confidence` rises in importance in the late sub-sample, consistent with increased narrative informativeness during the 2023–2025 deflationary window. Third, we restrict the sample to post-2015 (after the PBoC's communication regime reform) and re-estimate; the qualitative ordering of models is preserved.

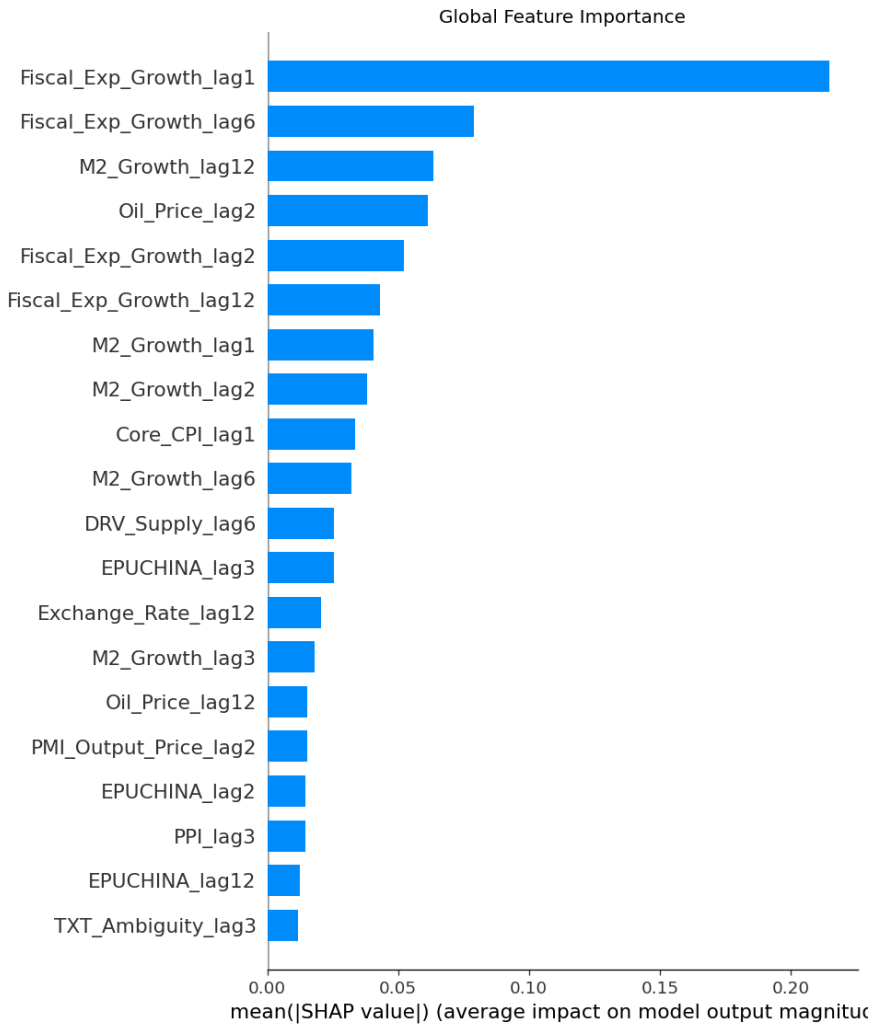

FIGURE 2. GLOBAL SHAP FEATURE IMPORTANCE, ENHANCED XGBOOST.

Note: Mean absolute SHAP value across the test window. Narrative features are prefixed `DRV_`, `INF_`, `POL_`, or `TXT_`; all others are macro variables.

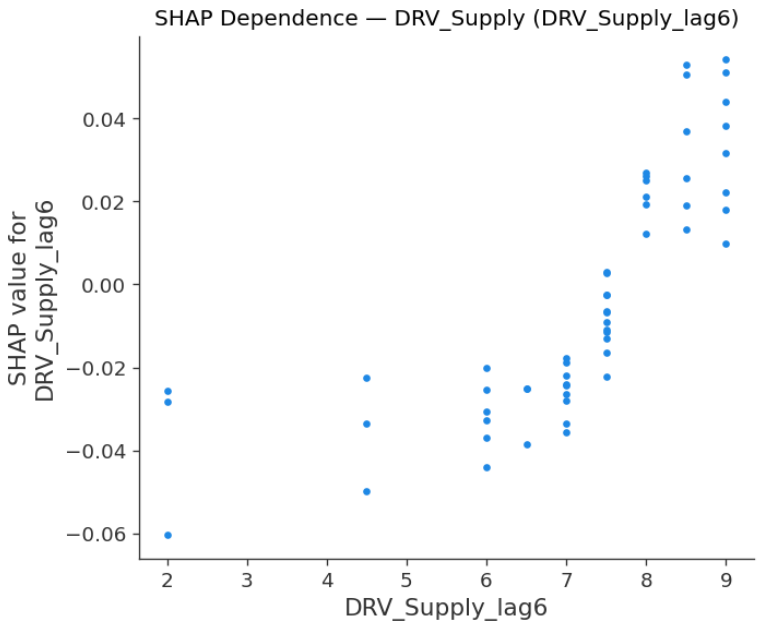

FIGURE 3. SHAP DEPENDENCE: DRV_SUPPLY_LAG6.

Note: Each point is a forecast date. Horizontal axis: value of the narrative feature. Vertical axis: SHAP contribution to $\hat{y}_t$.

VIII. Conclusion

This paper shows that an off-the-shelf large language model, given a well-specified narrative schema, can convert 25 years of PBoC *Monetary Policy Reports* into ten structured indicators that measurably improve CPI forecasts *provided* the downstream model can exploit interactions. The gains are economically but not statistically significant in our 56-observation test window, and are concentrated in non-linear tree ensembles rather than in penalized linear or factor models. Among the ten narrative dimensions, supply-driver attribution, textual confidence, and demand-driver attribution carry most of the predictive weight. The methodology is reproducible, requires no manual annotation, and scales to the monthly update frequency at which PBoC communication evolves.

Three limitations suggest directions for future work. First, our out-of-sample window is dominated by a single macroeconomic regime (post-COVID Chinese deflation); a longer out-of-sample horizon will become available mechanically as time passes. Second, we hold the LLM extraction fixed; an interesting extension is to randomise prompts and ensemble the resulting narrative features. Third, we do not model PBoC's *publication lag* (reports are released roughly one to two months after the quarter they discuss); correcting for the lag would generate a more conservative—but arguably more honest—baseline.

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Full LLM System Prompt

The system prompt passed to Gemini 3.0 Pro is reproduced in the replication package at `nlp/extract_inflation_narrative.py`. The prompt defines the role ("20-year macroeconomist and computational-linguistics specialist"), lists the ten dimensions with exact scale definitions, and instructs the model to reason economically rather than to keyword-match. Output is constrained to a JSON schema matching the ten columns of `data/nlp_features.csv`.

Additional Figures

Per-specification prediction paths (LASSO, Elastic Net, PLS, PCA+OLS, combination, LSTM, XGBoost) are provided at `outputs/cpi_growth_baseline/` and `outputs/cpi_growth_enhanced/`. Full SHAP dependence plots for each of the ten narrative dimensions (both XGBoost and random forest) are at `outputs/shap/{xgboost,random_forest}`